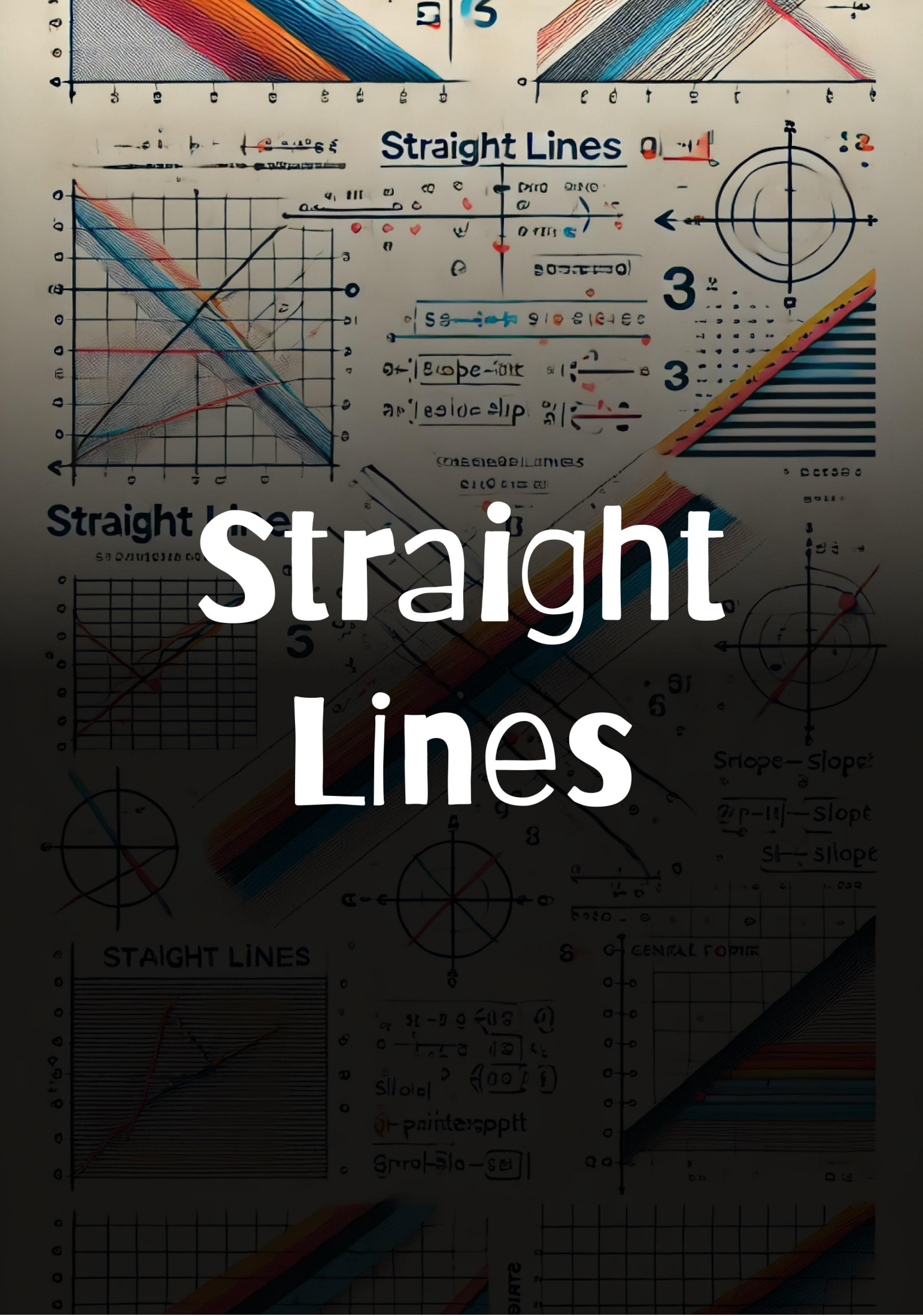
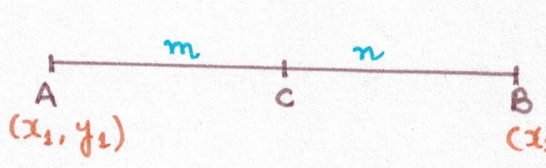


Straight Lines

STRAIGHT LINES

RATIO CONCEPT

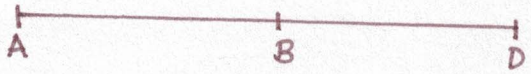
Internal Division



$$\frac{AC}{BC} = \frac{m}{n}$$

Co-ordinates of pt. C $= \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$

External Division



$$\frac{AD}{BD} = \frac{m}{n}$$

Co-ordinates of pt. D $= \left(\frac{mx_2 - mx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right)$

Ques: Find co-ordinates of the point M dividing P(2,3) and Q(6,7) in ratio of 1:3:

i) Internally

ii) Externally

Sol: i)



$m:n = 1:3$

Co-ordinates $= \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$

$= \left(\frac{6+6}{4}, \frac{7+9}{4} \right) = (3,4)$

ii)



Co-ordinates of M = $m:n = 1:-3$

$$= \left(\frac{2(-3)+6(1)}{-3+1} , \frac{3(-3)+7}{(-3)+1} \right)$$

$$= (0,1)$$

$A(x_1, y_1)$ and $B(x_2, y_2)$ are divided in ratio by the point $C(x_0, y_0)$ is -

$$x_0 - x_1 : x_2 - x_0$$

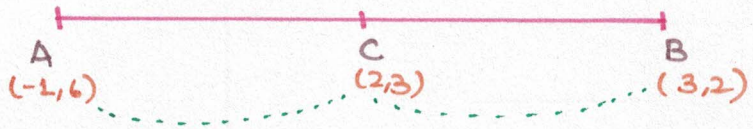
$$y_0 - y_1 : y_2 - y_0$$

If ratio is +ve ⟶ Internal division

If ratio is -ve ⟶ External division

Ques: Find ratio in which point (2,3) divides point A (1,-6) and B (3,2)

Sol:



Ratio = $2-(-1) : 3:2$

$= 3:1$

Internal Division

OR

Ratio = $3-(6) : 2-3$

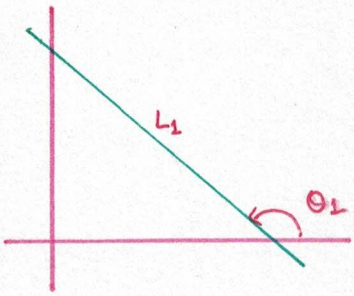
$= 3:1$

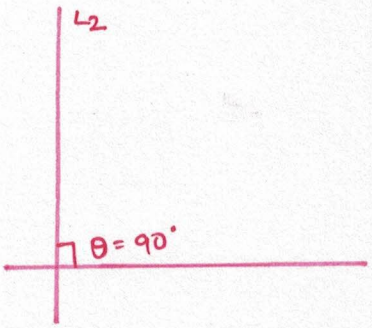
LINE

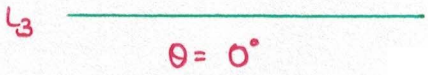
INCLINATION

Angle made by the line with x- axis in anticlockwise direction (θ)

$\theta \in [0, \pi)$

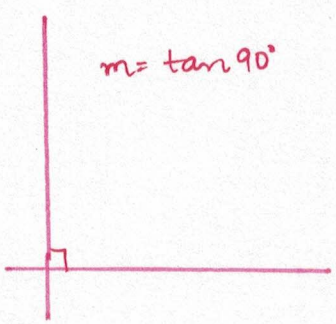


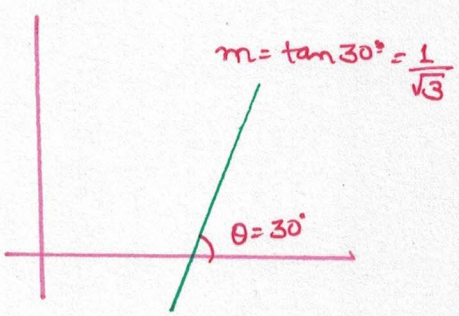




SLOPE OF A LINE

$$m = \tan\theta$$





Slope of a line passing through (x_1, y_1) and (x_2, y_2):

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Ques: Find slope of the line joining (1,2) and (7,5)

Sol: Slope = $m = \frac{3}{6} = \boxed{\frac{1}{2}}$

EQUATION OF STRAIGHT LINE:

- Requirements – (1) A point on line (2) Slope of line
- Let the line pass through the pt. (x_1, y_1)

(I) slope = $-a/b$, $ax + by = ax_1 + by_1$

(II) slope = a/b , $ax - by = ax_1 - by_1$

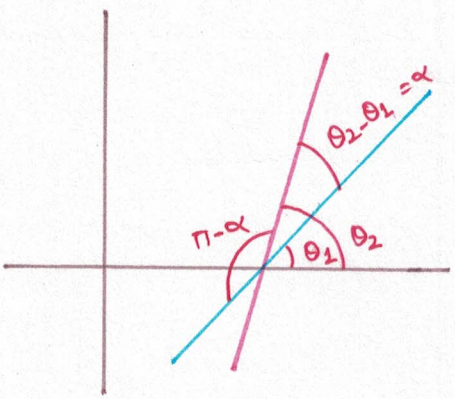
Ques: Slope = -2 , Point = (1, 2) , Find equation?

Sol: Slope = $\frac{-2}{1} = -\frac{a}{b}$

$2x+y = 2(1) + (1)(2)$

$\boxed{2x+y = 4}$

ANGLE BETWEEN TWO LINES



Let, acute angle between lines $= \alpha$

$\alpha = \theta_2 - \theta_1$

$\tan\alpha = \tan(\theta_2 - \theta_1)$

$\tan\alpha = \frac{\tan\theta_2 - \tan\theta_1}{1 + \tan\theta_2 . \tan\theta_1}$

$$\boxed{\tan\alpha = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|}$$ For acute $\angle$

$$\boxed{\text{Angle between two lines} = \tan^{-1}\left(\pm\frac{m_1 - m_2}{1 + m_1 m_2}\right)}$$

FINDING SLOPE OF A LINE

eg: $2x = 3y + 10$

$2x - 3y = 10$

$m = -\frac{\text{Coeff. of } x}{\text{Coeff. of } y} = -\frac{2}{-3} = \frac{2}{3}$

OR, $y = mx + c$

$3y = 2x + 10$

$y = \frac{2}{3}x + \frac{10}{3} \Rightarrow m = \frac{2}{3}$

REMEMBER

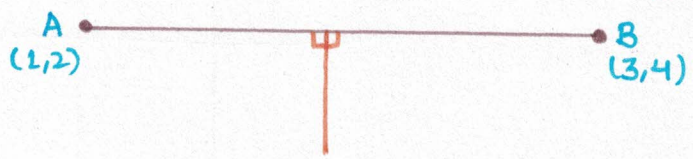
$$\tan\alpha = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

A) If lines are parallel, $m_1 = m_2$

b) If lines are $\perp$ to each other, $m_1 m_2 = -1$

Ques: Find slope of the line $\perp$ to the line joining (1,2), (3,4)

Sol:



If lines are $\perp$,

$$m_1 \cdot m_2 = -1$$

Slope of line AB,

$$m_1 = \frac{2}{2} = 1$$

$$m_2 = -1$$

PARALLEL AND PERPENDICULAR LINE STRUCTURE

$ax + by + c = 0$

	Parallel	Perpendicular
$ax + by = c$	$ax + by = \lambda$	$bx - ay = \lambda$
$ax - by = c$	$ax - by = \lambda$	$bx + ay = \lambda$

Ques: Line $\parallel$ to $5x = 7y + 6$

Ans: $5x - 7y = 6$

Line $\parallel \Rightarrow$ $5x - 7y = \lambda$

Ques: Line $\parallel$ to $x = -3y + 6$

Ans: $x + 3y = 6$

Line $\parallel \Rightarrow$ $x + 3y = \lambda$

Ques: Line $\parallel$ to $x = 0$

Ans: $x = \lambda$

Ques: Line ⊥ to $-3x+5y=7$

Ans: $3x-5y=-7$

$5x+3y=\lambda$

$m_1=\frac{3}{5}$, $m_2=\frac{-5}{3}$

$m_1 \cdot m_2=-1$ Verified

If three points given:

① If pts. are co-linear:-

(i) $AB+BC=AC$

(ii) $M_{AB}=M_{AC}$ or $M_{AB}=M_{BC}$

(iii) Ar. $(\Delta ABC)=0$

② If pts. are not co-linear ⇒ Points will frame Δ, find all sides AB, BC, CA

(i) $AB=BC=CA$ ⟶ Equilateral Δ

(ii) $AB=BC\neq CA$ ⟶ Isosceles Δ

(iii) $AB=BC\neq CA$, $CA^2=AB^2+BC^2$ ⟶ Right angle Isosceles Δ

③ $\boxed{l_1>l_2>l_3}$

(i) $l_1^2=l_2^2+l_3^2$ ⟶ Right angle Δ

(ii) $l_1^2>l_2^2+l_3^2$ ⟶ obtuse angle Δ

(iii) $l_1^2<l_2^2+l_3^2$ ⟶ Acute angle Δ

Ques: Find K if (K, K+1), (3,5), (K+2, 1) are co-linear

Sol: $\frac{5-K-1}{3-K}=\frac{1-K-1}{K+2-1}$

$K=\frac{1\pm\sqrt{33}}{2}$

If three lines given:-

① Find slope of the lines m_1, m_2, m_3

② $m_1 > m_2 > m_3$ (arrange in descending order)

③ Finding exact angles of triangle;

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2}, \quad \tan B = \frac{m_2 - m_3}{1 + m_2 m_3}$$

$$\tan C = \frac{m_3 - m_1}{1 + m_1 \cdot m_2}$$

④ (i) $A = B = C$ ⟶ Equilateral

(ii) $A = 90^\circ$ ⟶ Right Angle

(iii) $B = C = 45^\circ$, $A = 90^\circ$ ⟶ Right Angle Isosceles

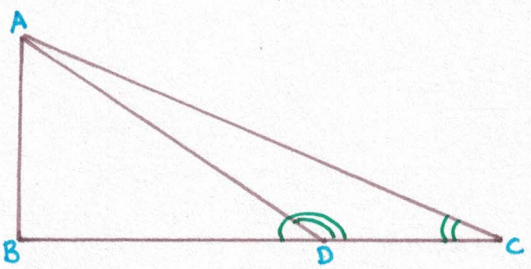
(iv) $B = C \neq A$ ⟶ Isosceles

(v) $\tan A < 0$ ⟶ Obtuse

(vi) $\tan A, \tan B, \tan C > 0$ ⟶ Acute

CENTROID

Median Median through A is a line joining A with the mid pt. of BC.

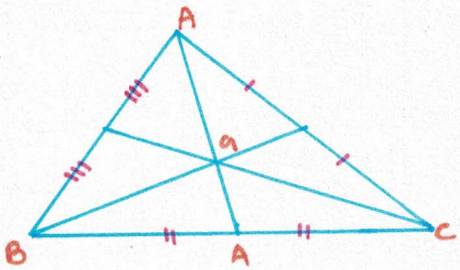


Median divides Δ in 2Δ having equal area

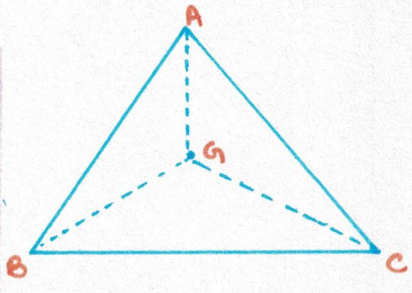


Ar (ΔABD) = Ar (ΔADC) = Area /2

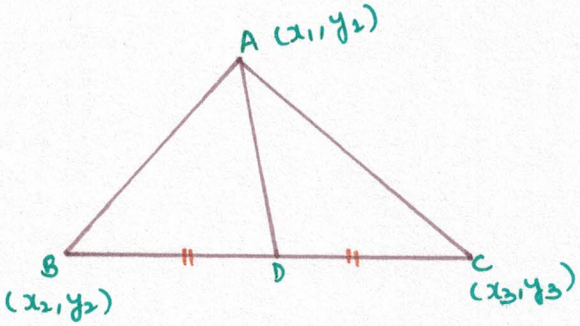
* All medians are concurrent, the point at which they meet is called **Centroid**



$Ar(\Delta ABC_1) = Ar(\Delta AGC) = Ar(\Delta CGB) = \frac{Area}{3}$

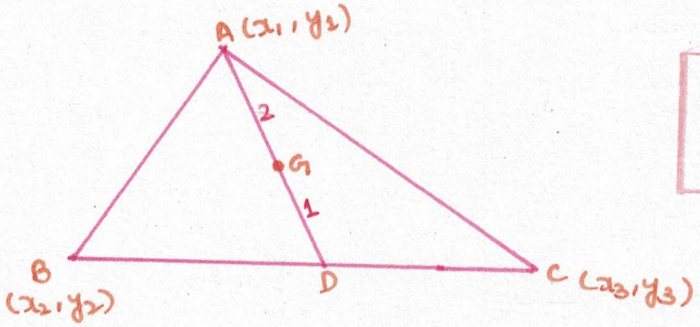


*



$$D = \frac{x_2+x_3}{2}, \frac{y_2+y_3}{2}$$

*



$$G = \frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}$$

Ques: If mid pts. of side BC, AB, CA are (2,0), (-2,0) and (2,2)

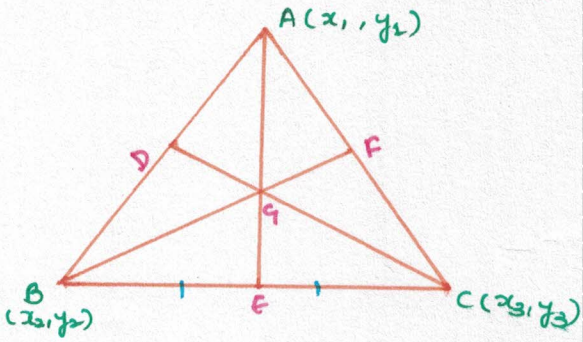
Find -

(i) All verticals

(ii) Equation of median through A

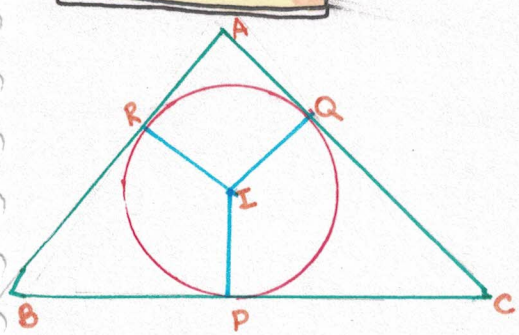
(iii) Length of median through C

(iv) Centroid G



Do it yourself!

INCENTER



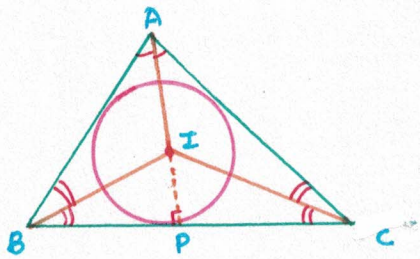
Circle which touches all sides of Δ is incircle, its center is **Incenter**

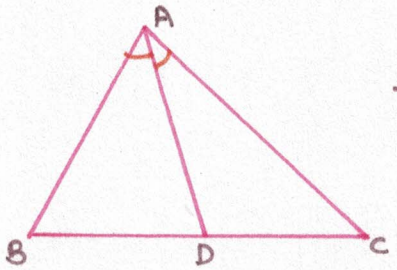
IP = IQ = IR = r = Area / Semiperimeter

$$r = \frac{\text{Area of } \Delta}{\text{Semi perimeter}}$$

$d(I, BC) = d(I, CA) = d(I, AB) = r$

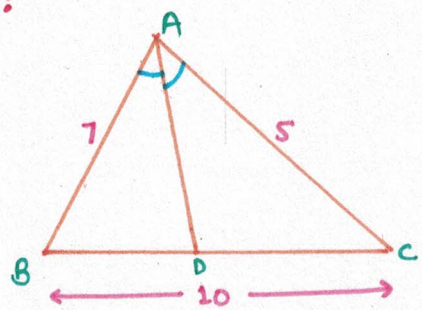
* Incenter is pt. of intersection of internal ∠ bisector





$$\frac{BD}{DC} = \frac{AB}{AC}$$

Ques:

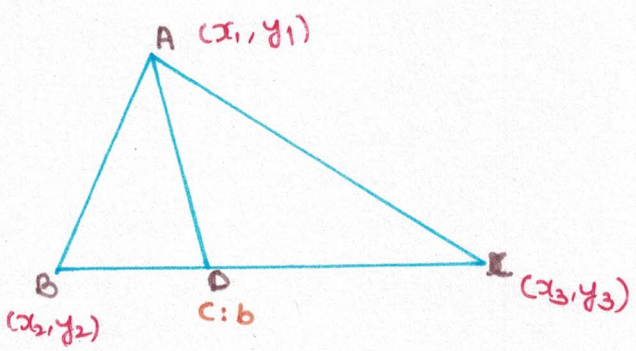


BD = ?

Ans: $\frac{x}{10-x} = \frac{7}{5}$

$x = 35/6$

*

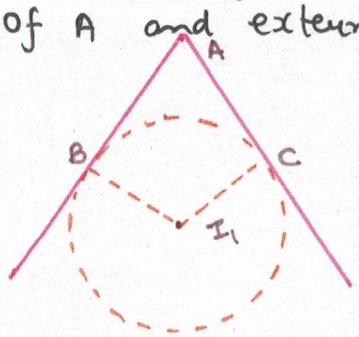


$$D = \left(\frac{bx_2 + cx_3}{b+c}, \frac{by_2 + cy_3}{b+c}\right)$$

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$$

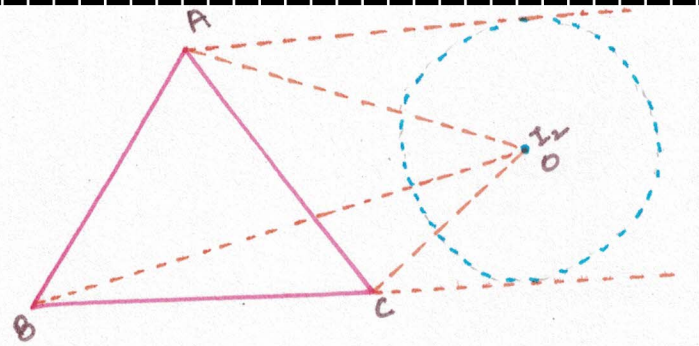
EXCENTRE

Excentre against A (Point of intersection of internal ∠ bisector of A and external ∠ bisector of B, C



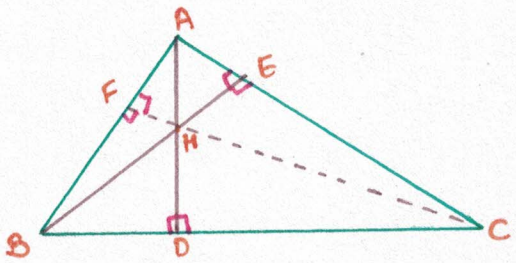
$$I_1 = \left(\frac{-ax_1 + bx_2 + cx_3}{-a+b+c}, \frac{-ay_1 + by_2 + cy_3}{-a+b+c}\right)$$

$$I_2 = \left(\frac{ax_1 - bx_2 + cx_3}{a-b+c}, \frac{ay_1 - by_2 + cy_3}{a-b+c}\right)$$

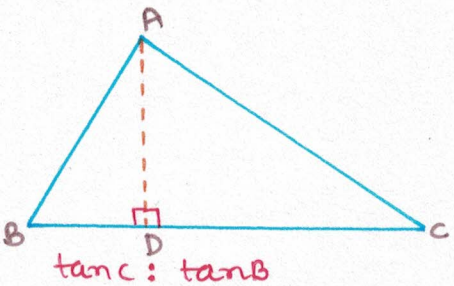


$$I_3 = \frac{ax_1 + bx_2 - cx_3}{a+b-c}, \frac{ay_1 + by_2 - c_3}{a+b-c}$$

ORTHOCENTER



* Altitude from A means line passing through A and ⊥ to BC.
* All altitudes of any Δ are concurrent (common point of intersection), point is known as the orthocenter



*

$$\frac{AD}{BD} = \tan B \Rightarrow BD = \frac{AD}{\tan B}$$

$$\frac{AD}{CD} = \tan C \Rightarrow CD = \frac{AD}{\tan C}$$

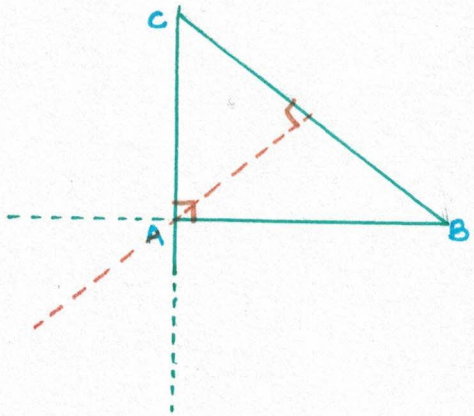
$$\frac{BD}{CD} = \frac{\tan C}{\tan B}$$

$$D = \frac{x_2 \tan B + x_3 \tan C}{\tan B + \tan C}, \frac{y_2 \tan B + y_3 \tan C}{\tan B + \tan C}$$

$$H = \frac{\{\tan B + \tan C\} \times D + \tan A \times A}{\tan A + \tan B + \tan C}$$

$$H = \frac{\tan A\, x_1 + \tan B x_2 + \tan C x_3}{\tan A + \tan B + \tan C}, \frac{\tan A y_1 + \tan B y_2 + \tan C y_3}{\tan A + \tan B + \tan C}$$

* For any right angles Δ, the orthocenter in vertex opposite to hypotenuse.



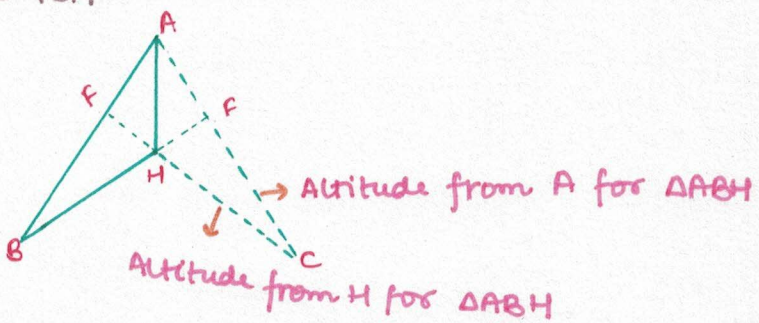
H is same as pt. A in Fig. $\boxed{H = A}$

* If H is orthocentre of ΔABC, then-

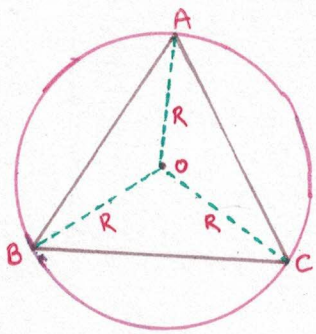
Orthocentre of ΔHBC = A

Orthocentre of ΔAHC = B

Orthocentre of ΔABH = C



CIRCUMCENTER



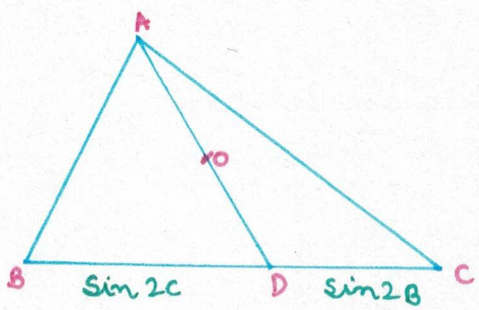
For any non co-linear pts. A, B, C a circle can pass through 3 pts. circle is called Circumcircle.

It's center → Circumcenter (O)

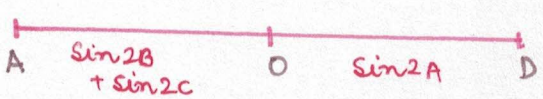
It's radius → Circumradius (R)

$\boxed{R = OA = OB = OC}$

★

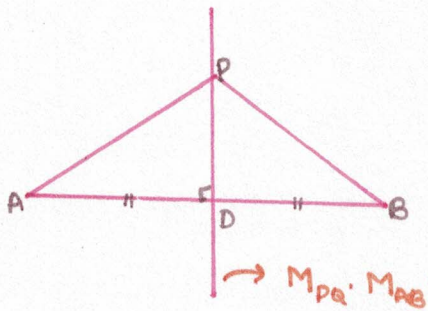


$$\boxed{D = \left(\frac{\sin 2B x_2 + \sin 2C x_3}{\sin 2B + \sin 2C}, \frac{\sin 2B y_1 + \sin 2C y_2}{\sin 2B + \sin 2C} \right)}$$



$$O = \left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C} \right)$$

PERPENDICULAR BISECTOR } of AB

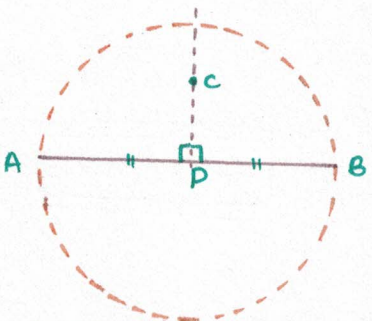


$\rightarrow M_{PQ} \cdot M_{AB} = -1$

- Passes through mid point of AB
- Will be ⊥ to line AB

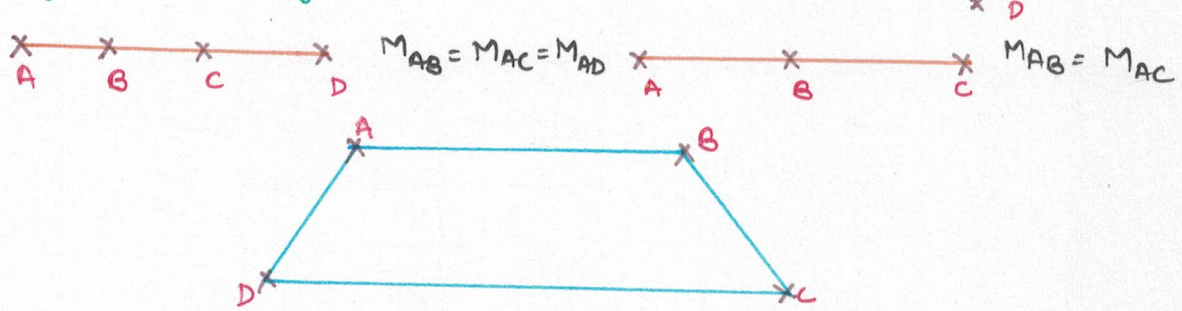
Locus of point (P) which means in such a way that for 2 fixed pts. A, B : PA = PB.

P is the ⊥ bisector of AB

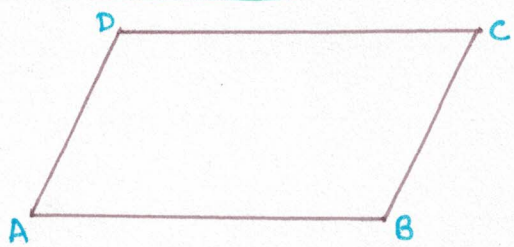


Perpendicular bisector of any chord passes through the center of circle.

If 4 Points given:



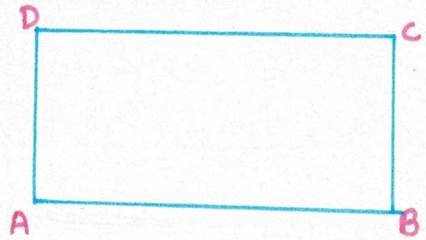
PARALLELOGRAM



Mid pt. of AC = Mid pt. of BD

$$\frac{A+C}{2} = \frac{B+D}{2}$$

RECTANGLE



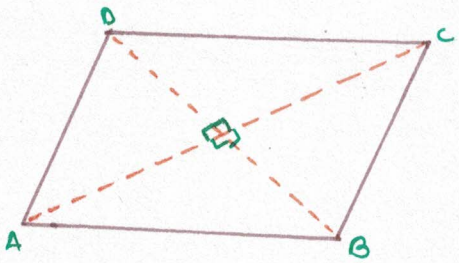
i) $A+C = B+D$

ii) $AD \perp AB$

$AB^2 + AD^2 = BD^2$

$M_{AB} \cdot M_{AD} = M_{DC} \cdot M_{CB}$

RHOMBUS

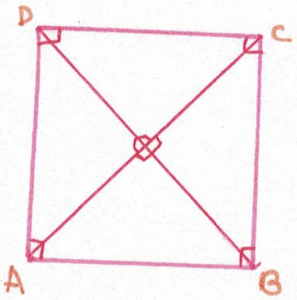


i) $A+C = B+D$

ii) $BD \perp AC$

$M_{BD} \cdot M_{AC} = -1$

SQUARE



i) $A+C = B+D$

ii) $AB \perp AD$

iii) $AC \perp BD$

iv) Diagonals bisect at 90°

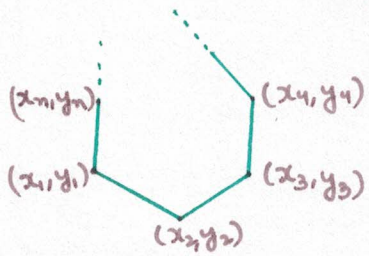
$M_{AC} \cdot M_{BD} = -1$

$AB^2 + AD^2 = BD^2$

OR

$M_{AD} \cdot M_{AB} = -1$

AREA OF POLYGON



Orthocentre's image with respect to any side of Δ, will lie on the circum-circle of the triangle

$$\text{Area} = \frac{1}{2} \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \\ x_3 & y_3 \\ \vdots & \vdots \\ x_n & y_n \end{vmatrix}$$

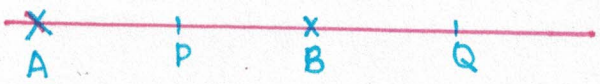
$$= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \begin{vmatrix} x_n & y_n \\ x_1 & y_1 \end{vmatrix}\right|$$

$$= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_1 & y_1 \end{vmatrix}\right| = \frac{1}{2}\left|\begin{vmatrix} x_2-x_1 & y_2-y_1 \\ x_3-x_1 & y_3-y_1 \end{vmatrix}\right|$$

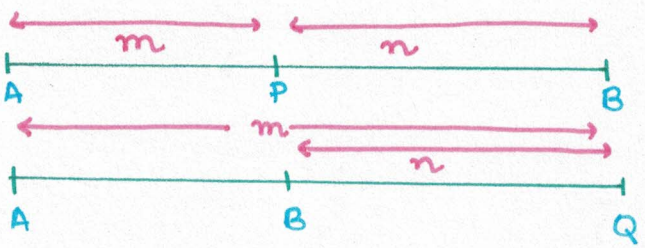
$$= \frac{1}{2}\left|\begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & y_2 \\ x_3 & y_3 \end{vmatrix} + \begin{vmatrix} x_3 & y_3 \\ x_4 & y_4 \end{vmatrix} + \begin{vmatrix} x_4 & y_4 \\ x_1 & y_1 \end{vmatrix}\right|$$

$$= \frac{1}{2}\left|\begin{vmatrix} x_3-x_1 & y_3-y_1 \\ x_4-x_2 & y_4-y_2 \end{vmatrix}\right|$$

HARMONIC CONJUGATE



(i) If P divides AB internally in m:n then its harmonic conjugate Q will be AB in m:n externally.



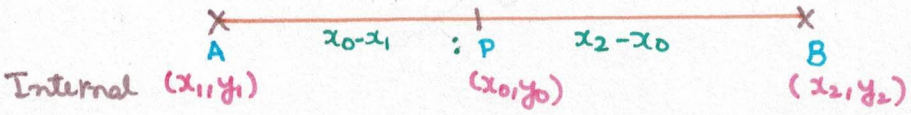
(ii) A, B, P, Q all points will be collinear

(iii) $$\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$$

Steps for finding harmonic conjugate:

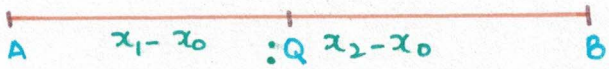
If $P(x_0, y_0)$, then its harmonic conjugate w.r.to $A(x_1, y_1)$, $B(x_2, y_2)$ is to find.

I)



Internal (x_1, y_1) (x_0, y_0) (x_2, y_2)

II)



∴ Apply ratio Formula

LOCUS

- Mathematical relation of variable point
- Equation in (x, y)
- Observatory Roots:

1) (a, a^2) ; (b, b^2) ; (c, c^2)
   x $y=x^2$

2) $\left(\frac{2}{a+1}, a\right)$; $\left(\frac{2}{b+1}, b\right)$; $\left(\frac{2}{c+1}, c\right)$
   x $y=2/x$

3) $\left(a, \frac{2}{a}\right)$, $\left(b, \frac{2}{b}\right)$, $\left(c, \frac{2}{c}\right)$
   x $y=x+1$

4) $(a, a+1)$, $(b, b+1)$, $(c, c+1)$
   x $y=x+1$

5) $\left(a, \frac{a^3}{a^2-1}\right)$, $\left(b, \frac{b^2}{b^2-1}\right)$, $y = \left(\frac{x^3}{x^2-1}\right)$

Parametric Locus:

Find x, find y and eliminate θ

Ques: Find locus of $(2\cos\theta, 3\sin\theta)$, θ is the parameter

Sol: $x = 2\cos\theta$, $y = 3\sin\theta$

$\cos\theta = \frac{x}{2}$, $\sin\theta = \frac{y}{3}$

$\cos^2\theta + \sin^2\theta = 1$

$\left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$

Ques: Find locus of the centroid of Δ where θ is parameter,

Vertices are $(1, 2)$, $(-1, 5)$, $(2\cos\theta, 2\sin\theta)$

Ans:

$$x = \frac{1 - 1 + 2\cos\theta}{3}$$

$$y = \frac{2 + 5 + 2\sin\theta}{3}$$

$$2\cos\theta = 3x$$

$$2\sin\theta = 3y - 7$$

$$\left(\frac{3x}{2}\right)^2 + \left(\frac{3y-7}{2}\right)^2 = 1$$

Locus: $9x^2 + (3y-7)^2 = 4$



EQUATION OF LINE IN VARIOUS FORMS



GENERAL FORM

$$Ax + By + C = 0$$

$$m = -A/B$$

Area made by Δ with the co-ordinate axis and line.

$y = 0$

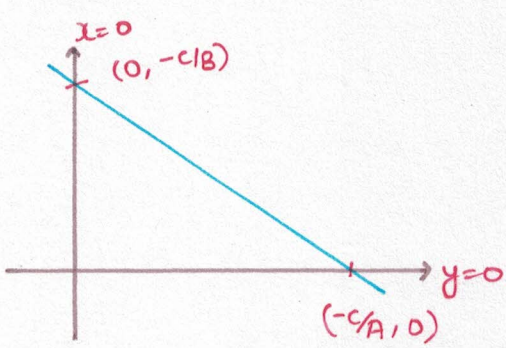
$$Ax + B \times 0 + C = 0$$

$$x = -\frac{C}{A}$$

$x = 0$

$$A \times 0 + By + C = 0$$

$$y = -\frac{C}{B}$$



$$\Delta = \frac{1}{2} \times \left| \left(\frac{-C}{A}\right) \cdot \left(\frac{-C}{B}\right) \right|$$

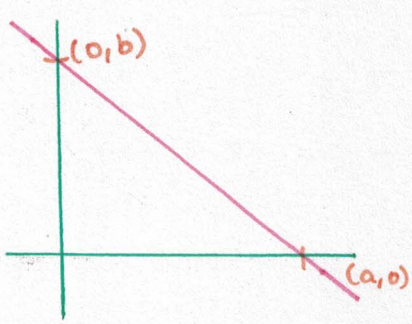
$$\Delta = \frac{1}{2} \times \left| \frac{C^2}{AB} \right|$$

INTERCEPT FORM

x- Intercept = a

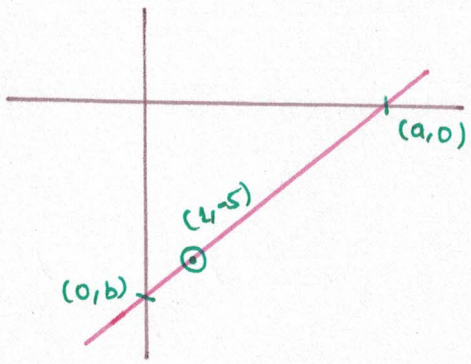
y- Intercept = b

$$\text{Line} = \frac{x}{a} + \frac{y}{b} = 1$$



Ques: Find equation of line whose intercept between axes is divided in 2:3 by (1, -5)

Ans: Let, equation be- $\frac{x}{a} + \frac{y}{b} = 1$



$\frac{3a}{5} = 1 \Rightarrow a = \frac{5}{3}$

$\frac{2b}{5} = -5 \Rightarrow b = -\frac{25}{2}$

NOTE:

i) Intercept on X-axis = Intercept on y-axis i.e $a = b$

$\frac{x}{a} + \frac{y}{b} = 1$

$x + y = a$ $\quad$ $m = -1$

ii) Length of intercept on x-axis = Length of intercept on y-axis

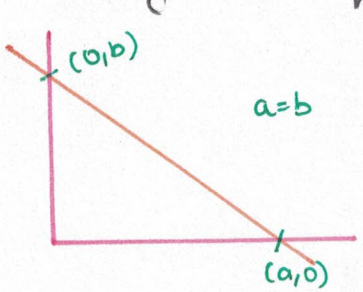
$m = \pm 1$

Ques: Equation of the line making equal intercept on co-ordinate axes and making a Δ with area = 8 units is -

i) $x + y = 4$
ii) $x - y = -4$
iii) $x + y = 2\sqrt{2}$
iv) $x - y = -2\sqrt{2}$

Sol: Area (8) = 8 units

Length is equal $\because m = \pm 1$



$x + y = a$ or $x - y = a$

Area = 12 units

$\frac{1}{2} \cdot |a^2| = 8$

$a = \pm 4$

$x \pm y = \pm 4$ } Equations (i) & (ii)

No. of possible lines = 4

CONVERSION OF GENERAL FORM TO INTERCEPT FORM

$Ax + By + C = 0$

$Ax + By = -C$

Divide by $-C$

$\frac{Ax}{-C} + \frac{By}{-C} = 1$

$\frac{x}{a} + \frac{y}{b} = 1$

$$\frac{x}{-C/A} + \frac{y}{-C/B} = 1$$

Ques: Convert in Intercept form:

a) $3x + 4y - 12 = 0$

$3x + 4y = 12$

$\frac{x}{4} + \frac{y}{3} = 1$

b) $2x - 5y + 7 = 0$

$2x - 5y = -7$

$\frac{x}{-7/2} + \frac{y}{7/5} = 1$

Intercept on x-axis $= -7/2$

y-axis $= 7/5$

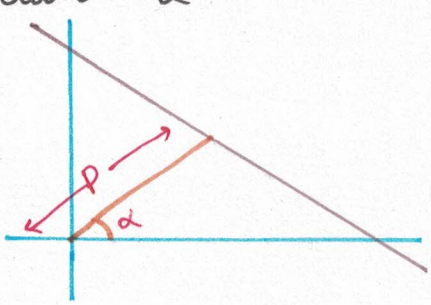
NORMAL FROM ORIGIN

Length of normal = P

Angle made by normal with ⊕ x-axis in anti-clockwise direction = α

$\alpha \in [0, 2\pi)$

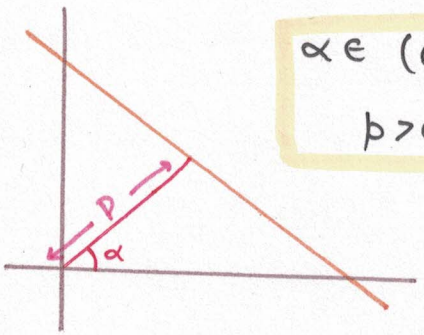
$P \geq 0$



$$\text{Line} = x\cos\alpha + y\sin\alpha = p$$

If $p \neq 0$,

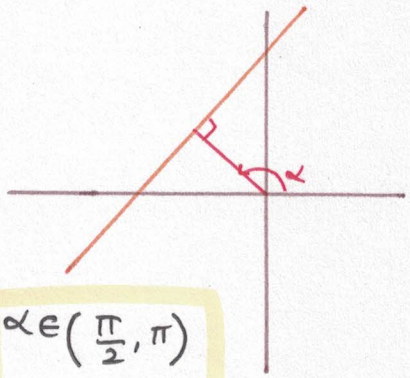
(i)



$\alpha \in (0, \frac{\pi}{2})$

$p > 0$

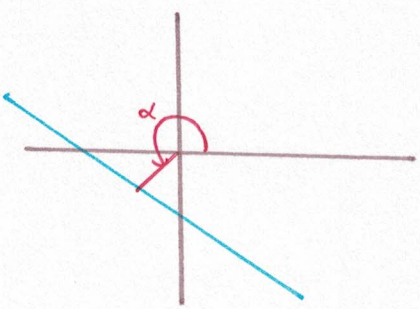
(ii)



$\alpha \in (\frac{\pi}{2}, \pi)$

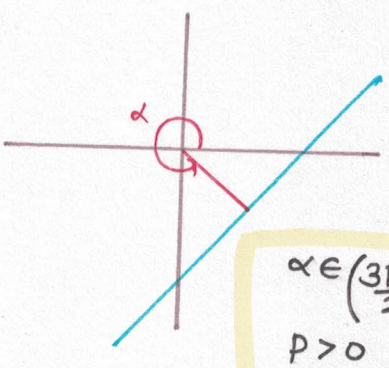
$p > 0$

(iii)



$\alpha \in (\pi, \frac{3\pi}{2})$

$p > 0$



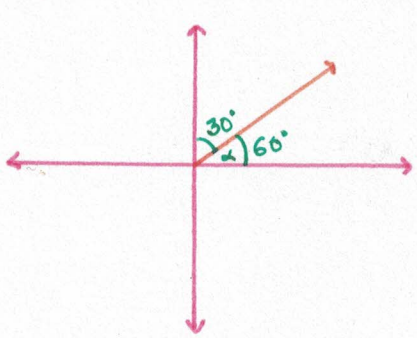
$\alpha \in (\frac{3\pi}{2}, 2\pi)$

$p > 0$



Ques: If normal makes an angle 30° with the +ve Y axis in clockwise direction, $p = 2$. Find eq^n of the line. Consider normal is from origin.

Sol:



$\angle = 60°$, $p = 2$

Line $= x \cos 60° + y \sin 60° = 1^2$

$\frac{x}{2} + \frac{y\sqrt{3}}{2} = 2$

$x + y\sqrt{3} = 4$

CONVERSION OF GENERAL FORM TO NORMAL FORM

$Ax + By + C = 0$

$Ax + By = -C$

$x \cos\alpha + y \sin\alpha = p$

$(p > 0)$

Divide by $\sqrt{A^2 + B^2}$

$$\frac{A}{\sqrt{A^2+B^2}}x + \frac{B}{\sqrt{A^2+B^2}}y = \frac{-C}{\sqrt{A^2+B^2}}$$

If it is +ve, then it is complete, but if it is -ve, multiply by $\ominus$ sign.

Ques: Convert in normal form.

(i) $x+y+10=0$

$x+y=-10$

Divide by $\sqrt{a^2+b^2}=\sqrt{2}$

$\Rightarrow \frac{x}{\sqrt{2}}+\frac{y}{\sqrt{2}}=\frac{-10}{\sqrt{2}}$

$\Rightarrow$ Multiply by $\ominus$

$\Rightarrow \frac{-x}{\sqrt{2}}-\frac{y}{\sqrt{2}}=\frac{10}{\sqrt{2}}$

$\Rightarrow -x\cos\frac{\pi}{4}-y\sin\frac{\pi}{4}=5\sqrt{2}$

$\Rightarrow \cos\alpha=\frac{-1}{\sqrt{2}}, \quad \sin\alpha=\frac{-1}{\sqrt{2}}$

$\Rightarrow \boxed{x\cos\frac{5\pi}{4}+y\sin\frac{5\pi}{4}=5\sqrt{2}}$

(ii) $x-\sqrt{3}y+2=0$

$\Rightarrow x-\sqrt{3}y=-2$

$\Rightarrow \frac{x}{2}-\frac{\sqrt{3}}{2}y=-2$

$\Rightarrow \frac{-1x}{2}+\frac{\sqrt{3}}{2}y=+2$

$\Rightarrow \cos\alpha=\ominus \quad \sin\alpha=\oplus$

$\Rightarrow \cos\left(\pi-\frac{\pi}{3}\right)=\cos\left(\frac{2\pi}{3}\right)$

$\Rightarrow \boxed{x\cos\frac{2\pi}{3}+y\sin\frac{2\pi}{3}=+2}$

$p=2, \quad \alpha=\frac{2\pi}{3}$

Point Slope Form

Point $=(x_1, y_1)$

Slope $=m$

$(y-y_1)=m(x-x_1)$

Ques: Find eqn of line if $m = \frac{2+\sqrt{3}-\sqrt{7}}{\sqrt{2}+5}$

Sol: Point is $(1+\sqrt{2}, 1)$

$$(y-y_1) = m(x-x_1)$$

$$(y-y_1) = \left(\frac{2+\sqrt{3}-\sqrt{7}}{\sqrt{2}+5}\right)(x-x_1)$$

$$(y-1) = \left(\frac{2+\sqrt{3}-\sqrt{7}}{\sqrt{2}+5}\right)(x-1-\sqrt{2})$$

Slope Intercept Form

y Intercept = c

Slope = m

Line $\Rightarrow$ $y = mx+c$

Point on line = $(0,c)$; slope = m

Ques: Find equation of line if slope = 2, y intercept is of length 3 below x-axis.

Sol: $2x-3=y$

POLAR FORM

Distance Form / Symmetric Form

If a line passes through (x_1, y_1) making angle θ with x-axis in the anti-clockwise direction ($m = \tan\theta$) then point (x,y) line which is at a distance $|r|$ from (x_1, y_1) can be given as-

(i) $\theta \in (0,\pi) - \{\frac{\pi}{2}\}$ $\quad \frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} = r$

(ii) r given: Points at r unit distance from (x_1, y_1)

i) $(x_1 + r\cos\theta, y_1 + r\sin\theta)$

ii) $(x_1 - r\cos\theta,\ y_1 - r\sin\theta)$

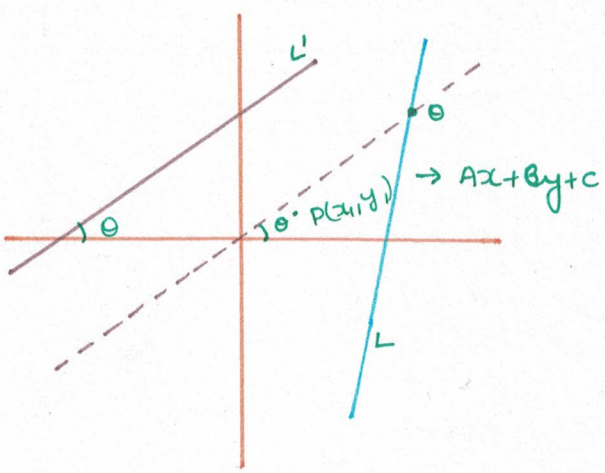
(iii) r not given $(x, y) = (x_1 + r\cos\theta,\ y_1 + r\sin\theta)$

(iv) Distance of (x_1, y_1) from $Ax+By+C$ measured parallel to a line with slope $(m = \tan\theta)$

OR, If P is (x_1, y_1), line p makes angle θ with x-axis in anti-clockwise direction cuts the line $Ax+By+C$.

Distance of a pt. (x_1, y_1) from the $Ax+By+C=0$ measured parallel along the line with slope $m = \tan\theta$:

$$\text{Distance} = \left| \frac{Ax_1 + By_1 + C}{A\cos\theta + B\sin\theta} \right|$$



Distance of P from L parallel to line PQ = ?

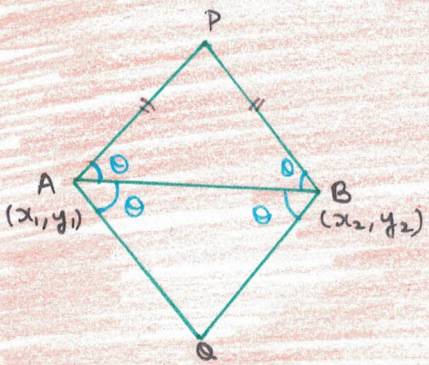
Let the distance PQ be 'r'.

Point on the line PQ at a distance r from P will be-

$(x_1 + r\cos\theta,\ y_1 + r\sin\theta)$

...REMEMBER...

- Isosceles Δ - Base (x_1, y_1), (x_2, y_2)



$$\frac{(x_1+x_2) \pm \tan\theta(y_1-y_2)}{2}, \qquad \frac{(y_1+y_2) \mp \tan\theta(x_1-x_2)}{2}$$

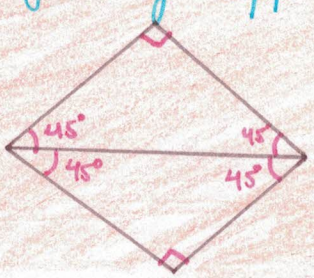
- Equilateral Triangle $(\theta = 60°)$

$$\frac{(x_1+x_2) \pm \sqrt{3}(y_1-y_2)}{2}, \quad \frac{(y_1+y_2) \mp \sqrt{3}(x_1-x_2)}{2}$$

- Right $\angle$ isosceles $(\theta = 45°)$

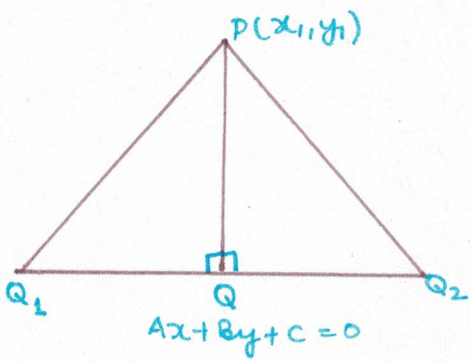
$$\frac{(x_1+x_2) \pm (y_1-y_2)}{2}, \quad \frac{(y_1+y_2) \mp (x_1-x_2)}{2}$$

- If A, B are diagonally opp. vertices of $\square$



Distance of (x_1, y_1) from $Ax + By + C = 0$

(Perpendicular distance or Minimum distance)



$$PQ = \left| \frac{Ax_1 + By_1 + C}{\sqrt{A^2+B^2}} \right|$$

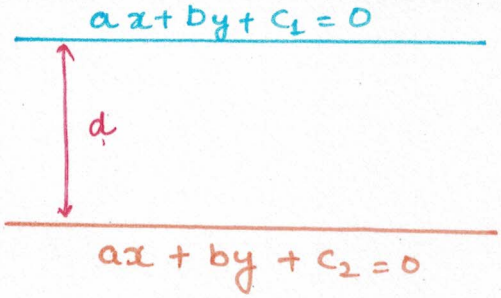
Proof: distance $= \left| \frac{Ax_1 + By_1 + C}{A\cos\theta + B\sin\theta} \right|$

For any function to be minimum.

Therefore, Denominator must be maximum

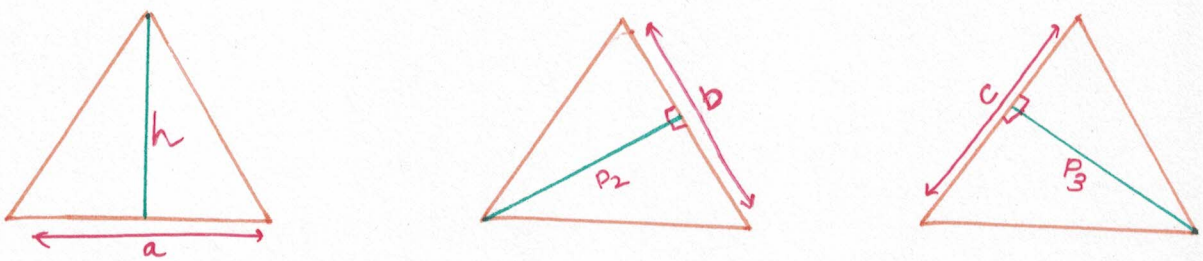
$$\therefore D = \left| \frac{Ax_1 + By_1 + C}{\sqrt{A^2+B^2}} \right|$$

Distance between two parallel lines



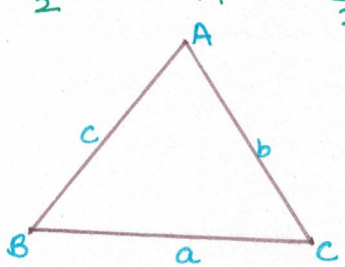
$$d = \frac{|c_1 - c_2|}{\sqrt{a^2+b^2}}$$

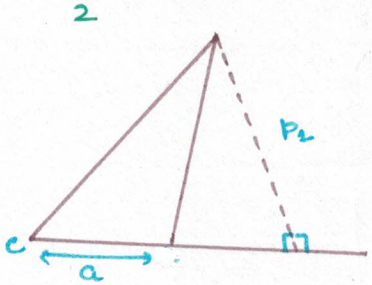
Area Of Triangle



$$\Delta = \frac{1}{2} \times a \times p_1 = \frac{1}{2} \times b \times p_2 = \frac{1}{2} \times c \times p_3$$

$$= \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B = \frac{1}{2} ab \sin C$$





$\Delta = \frac{1}{2} \times a \times p_1$

Position of points w.r.t. line

Ques: Check whether (-2, 1) and (3,2) lie on same side of $x-2y = 10$ or not.

Sol: Put the values of the pts. in the given equation of the line.

$\therefore \; x - 2y - 10 = 0$

If same sign occurs after putting the values, then the pt. lies on the same side of the line:

$(-2) - 2(1) - 10 \Rightarrow -14$

$3-(2)(2)-10 \Rightarrow -11$

$\therefore$ They lie on same side.

Ques: Check which of the following point lies on the same side as origin (0,0) to the line. $3x-y+2=0$

A) (1,2)

B) (-3,2)

C) (5,6)

D) (-1,1)

Ans: A) $3-2+2 \Rightarrow 3$ ✓

$0-0+2 \Rightarrow 2$ ✓

B) $-9-2+2 \Rightarrow -9$ ×

$0-0+2 \Rightarrow 2$ ×

(A, C)

C) $15-6+2 \Rightarrow 11$ ✓

$0-0+2 \Rightarrow 2$ ✓

D) $-3-1+2 \Rightarrow -2$ ×

$3-1+2 \Rightarrow 2$ ×

Note:

If $A(x_1, y_1)$; $B(x_2, y_2)$ and line $ax+by+c=0$; A, B lies on same section of the line if L_{1A}, L_{2B} have same slope.

A, B doesn't lie on the line if L_{1A}, L_{2B} have different signs

$L_{1A} = ax_1+by_1+c$; $L_{1B} = ax_2+by_2+c$

Note:

L_1 is an equation of line. If pts. A, B lie on:

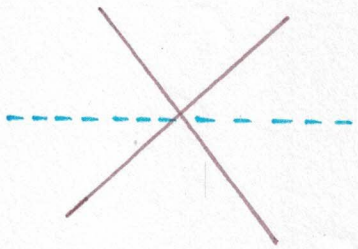
(i) Same side of $L_1 \Rightarrow L_{1A} \cdot L_{1B} > 0$

(ii) Same side of $L_1 \Rightarrow L_{1A} \cdot L_{1B} < 0$

FAMILY OF LINES

CONCURRENT LINES ⟶ Lines having same point of intersection

If lines are ||, no solution,
They can't be concurrent



METHOD-I

$$a_1x + b_1y + c_1 = 0$$
$$a_2x + b_2y + c_2 = 0$$
$$a_3x + b_3y + c_3 = 0$$

(i) Find P.O.I.
(ii) Substitute that pt. on line

METHOD-II

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

Ques: Find λ if $2x+3y=5$, $7x+6y+10=0$, $\lambda x+y-3=0$ are concurrent.

Sol: The determinant will be zero

$$\begin{vmatrix} 2 & 3 & -5 \\ 7 & 6 & 10 \\ \lambda & 1 & -3 \end{vmatrix} = 0$$

$$\Rightarrow -56+30\lambda+63-35+30\lambda = 0$$

$$\Rightarrow 60\lambda = 28$$

$$\Rightarrow \boxed{\lambda = \frac{7}{15}}$$

Ques: If lines $x+\lambda y+10=0$, $x+\mu y+14=0$, $x+\gamma y+18=0$, find the constraint if the lines are in concurrent, then λ, μ, γ are -

(A) in AP (B) GP (C) HP (D) None

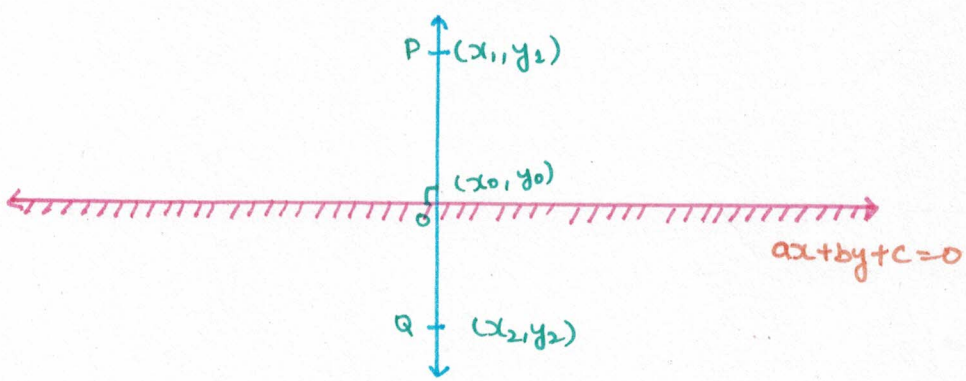
Sol:
$$\begin{vmatrix} 1 & \lambda & 10 \\ 1 & \mu & 14 \\ 1 & \gamma & 18 \end{vmatrix} = 8\mu - 4\gamma - 4\lambda = 0$$

$$= 2\mu - \gamma + \gamma$$

$$= 2\mu - \lambda + \gamma$$

λ, μ, γ are in A.P.

Foot Of Perpendicular Image



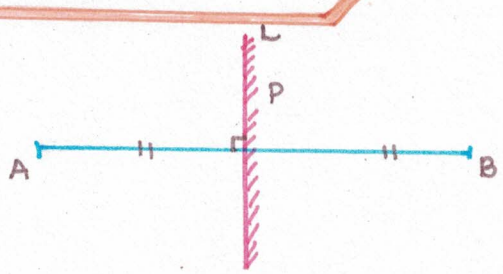
Foot of Perpendicular:

$$\frac{x_0 - x_1}{a} = \frac{y_0 - y_1}{b} = \frac{-(ax_1 + by_1 + c)}{a^2 + b^2}$$

Image (x_2, y_2):

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b} = \frac{-2(ax_1 + by_1 + c)}{a^2 + b^2}$$

Perpendicular Bisector



Locus of a point which moves in such a way that, distance from two points are equal.

(i) $PA = PB$

(ii) $L \perp AB$, L passes through mid pt. of AB

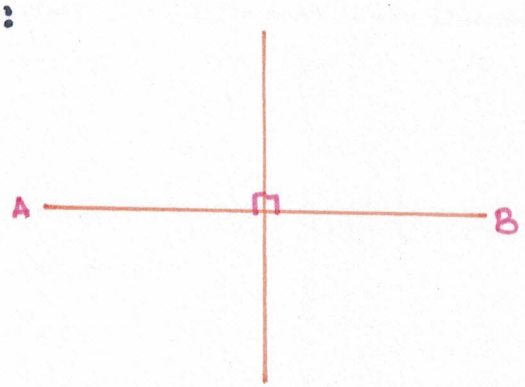
(iii) If L is $\perp$ bisector of AB, then-

Image of A in $L = B$

Image of B in $L = A$

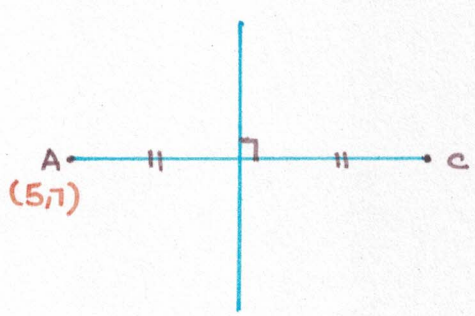
Ques: In ΔABC, the eq^n of right bisectors of side AB and AC are $x+y=0$ and $x-y=0$ respectively. If $A = (5,7)$. Find eq^n of BC.

Ans:



As $x+y=0$ is $\perp$ bisector of AB

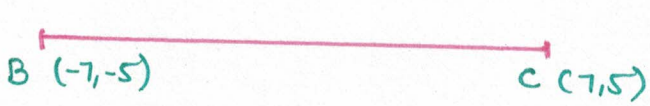
B is image of A w.r.t. $x+y=0$



C is the image of A w.r.t. $\perp$ bisector $x-y=0$

$C = (7,5)$

Line BC;



$M = \frac{10}{14} = \frac{5}{7}$, pt. $= (7,5)$

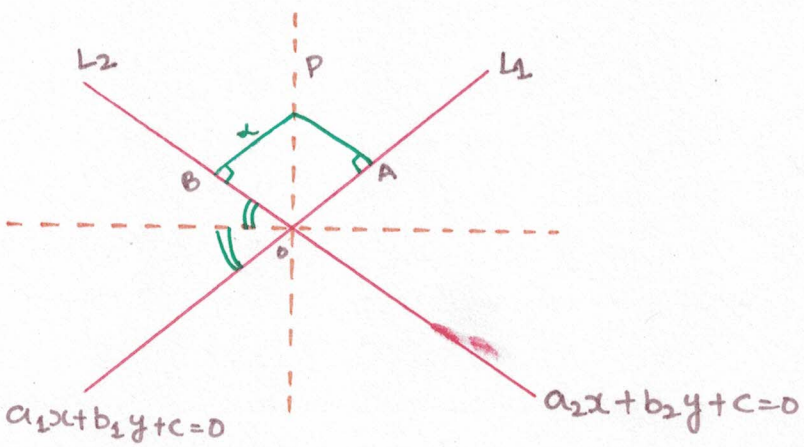
$5x - 7y = 0$

ANGLE BISECTOR:

Locus of a point which moves in such a way that its distance from two lines:

$$\left|\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}\right| = \left|\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}\right|$$

$$\left(\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}\right) \pm \left(\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}\right) = 0$$



Ques: Find equation of angle bisectors of the line $x+7y-8=0$ and $x+y-2=0$, also check which is acute $\angle$ bisector and which is obtuse.

Sol: $$\left(\frac{x+7y-8}{\sqrt{50}}\right) \pm \left(\frac{x+y-2}{\sqrt{2}}\right) = 0$$

Case-I +ve:

$$\Rightarrow \frac{x+7y-8}{\sqrt{50}} + \frac{x+y-2}{\sqrt{2}} = 0$$

$$\Rightarrow (x+7y-8)\sqrt{2} + (x+y-2)\sqrt{50} = 0$$

$$\Rightarrow \sqrt{2}x + 7\sqrt{2}y - 8\sqrt{2} + 5\sqrt{2}x + 5\sqrt{2}y - 10\sqrt{2}$$

$$\Rightarrow 6\sqrt{2} + 12\sqrt{2}y - 18\sqrt{2} = 0$$

$$\Rightarrow \boxed{x+2y-3=0} \quad \text{— (1)}$$

Case-II -ve

$$\Rightarrow \frac{x+7y-8}{5\sqrt{2}} - \left(\frac{x+y-2}{\sqrt{2}}\right) = 0$$

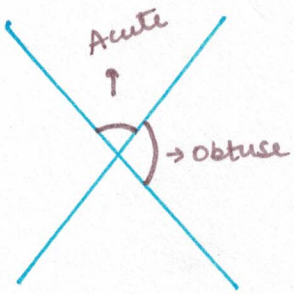
$$\Rightarrow x\sqrt{2} + 7\sqrt{2}y - 8\sqrt{2} - 5\sqrt{2}x - 5\sqrt{2}y + 10\sqrt{2} \;/\; (5\sqrt{2})\sqrt{2} = 0$$

$\Rightarrow -4\sqrt{2}x + 2\sqrt{2}y + 2\sqrt{2} = 0$

$\Rightarrow \boxed{2x - y + 1 = 0}$ — (2)

$x + 2y - 3 = 0$ will be acute angle bisector

$2x - y + 1 = 0$ will be obtuse angle bisector

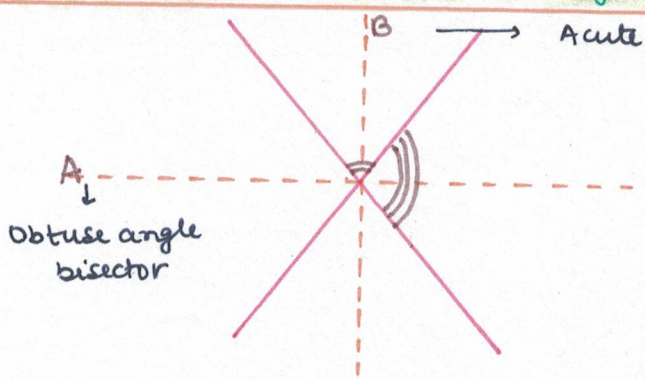


Let line $= x + y - 2 = 0 \Rightarrow m = -1$

$x + 2y - 3 = 0 \Rightarrow m = -1/2$

$\therefore$ This angle is acute angle bisector

For verifying whether A, B is acute angle or obtuse angled



Any L_1 and any angle bisector

Angle between them $\alpha < 45°$, $|\tan\alpha| < 1$ } Angle bisector is acute angle bisector

Angle between them $\alpha > 45°$, $|\tan\alpha| > 1$ } Angle bisector is obtuse angle bisector

Note: Directly finding acute $\angle$ bisector and obtuse $\angle$ bisector

Acute angle bisector $\rightarrow$ $\dfrac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}}$ $\dfrac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$

(Same as sign of $a_1a_2 + b_1b_2$)

Obtuse angle bisector $\rightarrow$ $\dfrac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}}$ $\dfrac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$

(opposite sign of $a_1a_2 + b_1b_2$)

Ques: Find acute angle bisector of $x+7y-8=0$ and $x+y-2=0$

Ans: $\frac{x+7y-8}{\sqrt{50}}$ (Sign of $1\times1+7\times1$) $\frac{x+y-2}{\sqrt{1^2+1^2}}=0$

$$\frac{x+7y-8}{\sqrt{50}}+\frac{x+y-2}{\sqrt{2}}=0$$

Acute angle bisector is $x+2y-3=0$

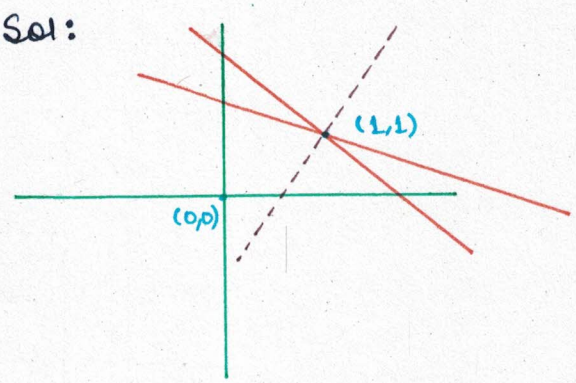
Note:2 Angle bisector containing a point:

Convert the line in general form. $L_1; L_2$ bisector of angle containing point P

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} \text{ (opp sign of } L_1p\cdot L_2p) \frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}=0$$

Ques: Find bisector of angle of lines $x+7y-8=0$ and $x+y-2=0$, which contains the origin.

Sol:



$$\left(\frac{x+7y-8}{\sqrt{50}}\right)\pm\left(\frac{x+y-2}{\sqrt{2}}\right)=0$$

$x+2y-3=0$

$2x-y-1=0$

$\therefore$ The correct answer is $2x-y=1$

OR

$P=(0,0)$, $L_1=x+7y-8=0$, $L_2=x+y-2=0$

$L_1p\cdot L_2p=(-8)(-2)=16$

$$\left(\frac{x+7y-8}{\sqrt{50}}\right)-\left(\frac{x+y-2}{\sqrt{2}}\right)=0$$

Answer is $2x-y-1=0$

Ques: Find equation of bisector of $<$ between the line $21x+23y=7$ and $23x+21y+2=0$, which contains $(1,1)$.

Ans: $L_1p = 21+23-7$

$L_2p = 23+21+2$ } +ve

$$\therefore \frac{21x+23y-7}{\sqrt{970}} - \frac{23x+21y+2}{\sqrt{970}}$$

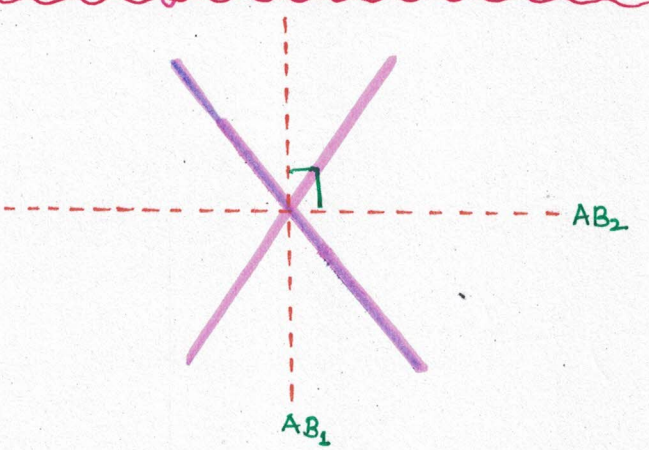
$$= -2x+2y-9=0$$

$$= 2x-2y+9=0 \quad \} \text{ Answer}$$

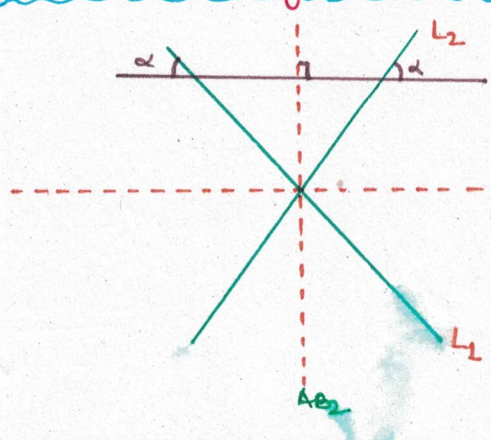
Note3: Bisector not containing a pt. P :-

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} \quad (\text{Same sign as of } L_1P \cdot L_2P) \quad \frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}} = 0$$

Note4: Angle between angle bisectors of two lines = 90°



Note5: Line equally inclined with two given lines will be parallel / perpendicular to angle bisectors of two lines.



Remember by X and y axis

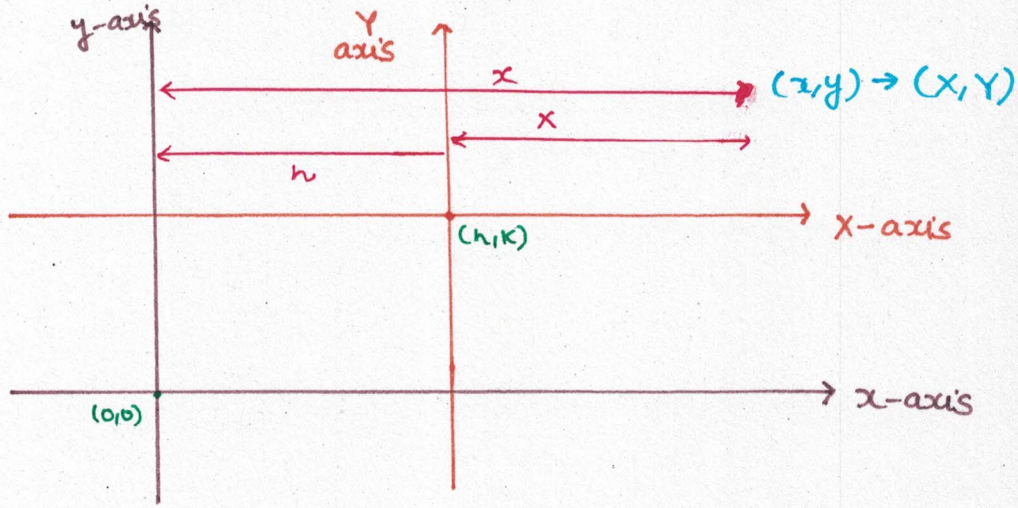
SHIFT OF ORIGIN

$$x = X + h$$
$$y = Y + K$$

$(x, y) \rightarrow$ Old co-ordinate

$(X, Y) \rightarrow$ New co-ordinate

$(h, K) \rightarrow$ Point where origin is shifted



ROTATION OF AXES

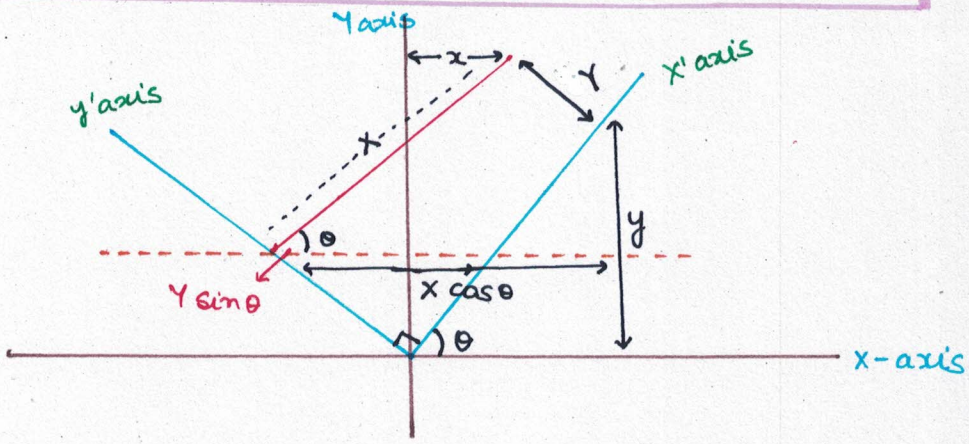
$$x = X\cos\theta - Y\sin\theta$$

$$y = X\sin\theta + Y\cos\theta$$

	X	Y
x	$\cos\theta$	$-\sin\theta$
y	$\sin\theta$	$\cos\theta$

$x = X\cos\theta + y(-\sin\theta)$ $\quad$ $y = X\sin\theta + y\cos\theta$

$X = x\cos\theta + y\sin\theta$ $\quad$ $Y = x(-\sin\theta) + y\cos\theta$



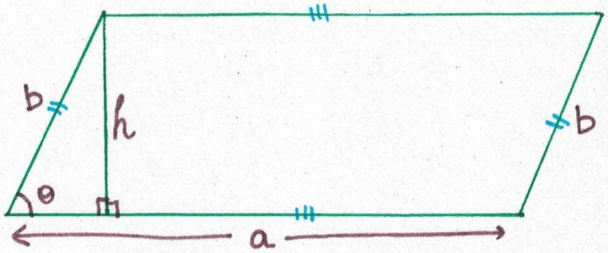
Ques: Identify the shape:

$12x - 5y - 10 = 0$, $5x + 12y + 2 = 0$, $12x - 5y - 20 = 0$, $5x + 12y - 8 = 0$

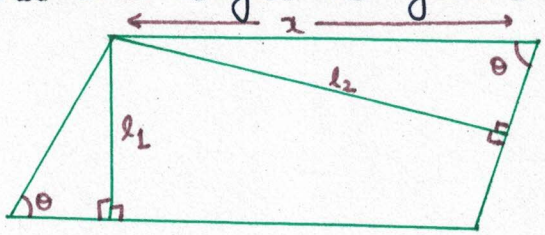
Ans: It's a square

AREA OF A PARALLELOGRAM

Area of ▱ = $a \times h = ab\sin\theta$



If two lines and one angle is given.

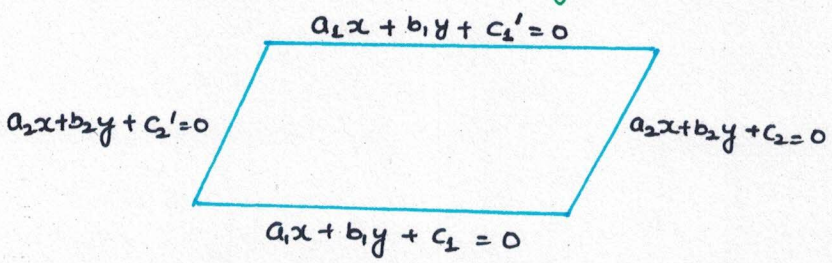


$$\frac{l_2}{x} = \sin\theta$$

$$x = \frac{l_2}{\sin\theta}$$

$$\text{Area of } \square = \frac{l_1 l_2}{\sin\theta}$$

Area of a Quadrilateral Formed by:-



$$\text{Area} = \frac{|c_1 - c_1'|\,|c_2 - c_2'|}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

Ques: Find the area of quadrilateral formed by the lines,
$2x - 3y - 1 = 0$, $2x + 3y + 2 = 0$, $2x - 3y + 1 = 0$, $2x + 3y = 0$

Ans:

$$\text{Area} = \frac{2 \cdot 2}{\begin{vmatrix} 2 & +3 \\ 2 & -3 \end{vmatrix}}$$

$$= \frac{4}{|6+6|}$$

$$= \frac{4}{12}$$

$$= \frac{1}{3} \text{ Sq. units}$$